

NFL Architecture Review Board Implementation Specification

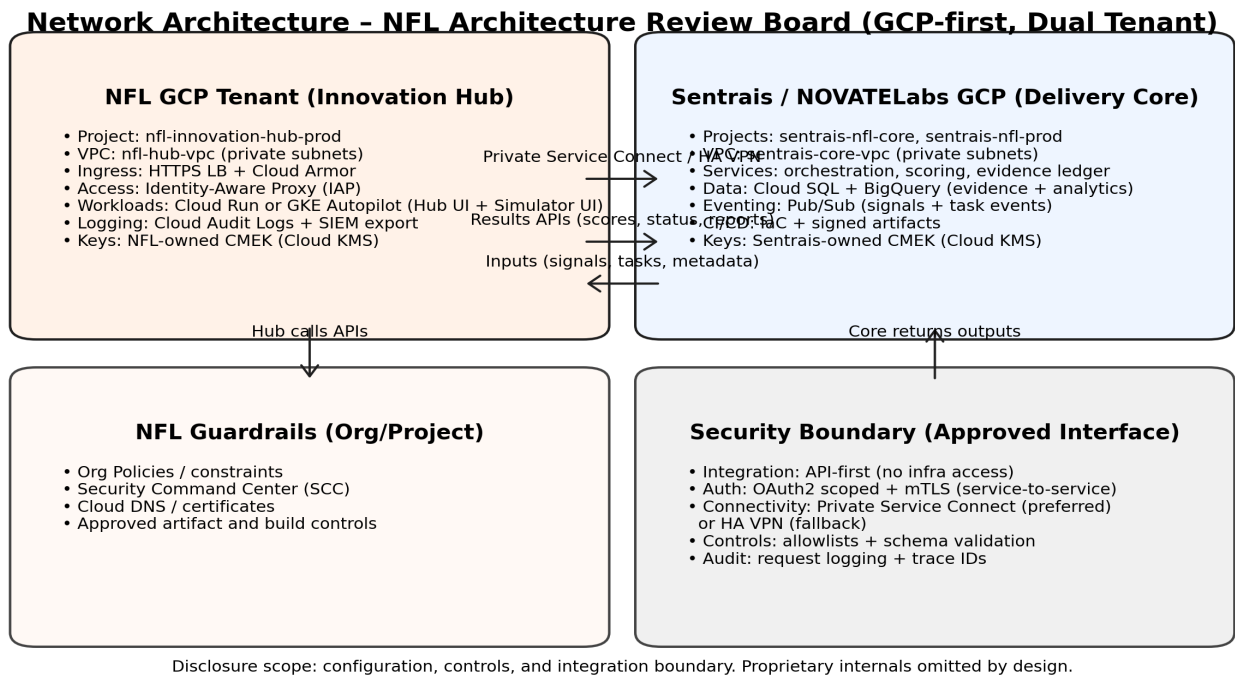
Scope: GCP-first deployment for the NFL Innovation Hub tenant and secure integration to Sentrais-managed GCP delivery environments.

1. Objectives

- Establish a secure GCP configuration for the NFL Innovation Hub.
- Enable controlled integration to Sentrais services for readiness, scoring, orchestration, and reporting.
- Enforce tenant isolation, least privilege, encryption, and auditability.

2. Architecture Summary

The NFL Innovation Hub operates in an NFL-owned GCP project and VPC. Sentrais operates a separate GCP environment executing proprietary services. The environments integrate via private connectivity and API-based contracts. No shared administrative access is required.



3. NFL GCP Configuration (Hub Tenant)

Projects: nfl-innovation-hub-prod (and optional nfl-innovation-hub-dev).

Network: nfl-hub-vpc with private subnets only; controlled egress.

Ingress: External HTTPS Load Balancer + Cloud Armor; IAP enforced for user access.

Workloads: Cloud Run or GKE Autopilot for Hub UI and simulator UI.

Encryption: NFL-managed CMEK (Cloud KMS) applied per NFL standard.

Logging: Cloud Audit Logs enabled; log sinks exported to NFL security monitoring.

4. Integration Configuration (How the Hub Connects to Sentrais)

Connectivity: Private Service Connect (preferred) with allowlisted endpoints; HA VPN fallback if required.

Auth: Scoped OAuth2 principals and service identities; mTLS for service-to-service requests where applicable.

Boundary: API-only integration; no direct database or infrastructure access across tenants.

Data Contracts: Schema validation for requests; standardized outputs (scores, status, evidence-linked reports, simulator results).

5. Responsibilities (RACI)

Area	NFL	Sentrais / NOVATELabs
Hub tenant IAM, keys, logging	A/R	C
Hub UI hosting and user access policy	A/R	C
Core execution services (orchestration, scoring, evidence)	A/R	A/R
Integration boundary controls (connectivity, auth, APIs)	A/R	A/R
Incident coordination and audit support	A/R	A/R

6. ARB Decisions Required

- 1) Confirm Hub hosting standard (Cloud Run vs GKE Autopilot).
- 2) Confirm connectivity preference (Private Service Connect vs HA VPN).
- 3) Confirm log export destination, retention, and monitoring thresholds.
- 4) Confirm CMEK scope requirements for Hub tenant services.
- 5) Confirm permitted data fields for integration contracts.